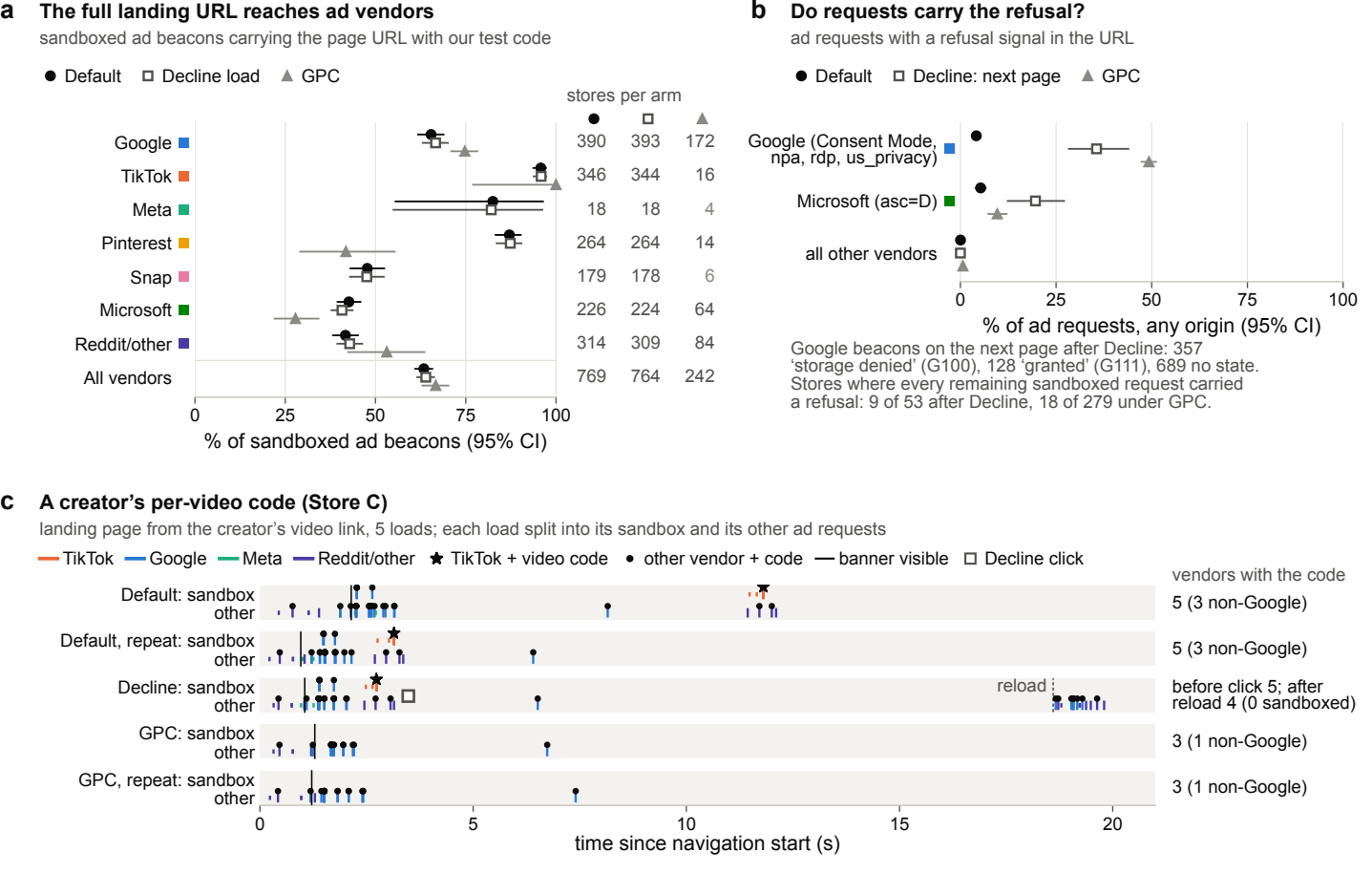

a
The full landing URL reaches ad vendors
sandboxed ad beacons carrying the page URL with our test code
Default
Decline load
GPC
stores per arm
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
All vendors
390 393 172
346 344 16
18 18 4
264 264 14
179 178 6
226 224 64
314 309 84
769 764 242
0
25
50
75
100
% of sandboxed ad beacons (95% CI)
b
Do requests carry the refusal?
ad requests with a refusal signal in the URL
Default
Decline: next page
GPC
Google (Consent Mode, npa, rdp, us_privacy)
Microsoft (asc=D)
all other vendors
0
25
50
75
100
% of ad requests, any origin (95% CI)
Google beacons on the next page after Decline: 357 'storage denied' (G100), 128 'granted' (G111), 689 no state.
Stores where every remaining sandboxed request carried a refusal: 9 of 53 after Decline, 18 of 279 under GPC.
c
A creator's per-video code (Store C)
landing page from the creator's video link, 5 loads; each load split into its sandbox and its other ad requests
TikTok
Google
Meta
Reddit/other
TikTok + video code
other vendor + code
banner visible
Decline click
vendors with the code
Default: sandbox
other
5 (3 non-Google)
Default, repeat: sandbox
other
5 (3 non-Google)
Decline: sandbox
other
reload
before click 5; after reload 4 (0 sandboxed)
GPC: sandbox
other
3 (1 non-Google)
GPC, repeat: sandbox
other
3 (1 non-Google)
0
5
10
15
20
time since navigation start (s)